Quantum and Quantum-Inspired Graph Neural Networks for Dynamic Wireless Network Topologies: A Tutorial and Survey with a Reproducible LEO Case Study

Phuc Hao Do

***Abstract*—Future wireless networks, from low Earth orbit (LEO) mega-constellations to reconfigurable access fabrics, are time-varying graphs, and graph learning is becoming a central tool for managing them. At the same time, quantum and quantum-inspired learning is promoted as the next step for such networks, yet the literature rarely makes precise what a quantum model changes, or how to tell a real advantage from an artifact of evaluation. This tutorial offers a unifying answer. We show that classical graph neural networks, graph diffusion, and continuous-time quantum walks are the same object, a propagation operator on the graph Laplacian, so that the only difference between them is the spreading physics they encode: diffusive for the heat kernel, ballistic for the quantum walk. From this view we derive a param-matched, quantum-inspired graph neural network layer that runs on classical hardware today, and we teach how to evaluate it honestly. Using a fully reproducible case study on a dynamic LEO graph, we show that global propagation gives a large and robust gain over local message passing, while among global operators the quantum-inspired and classical kernels trade places *with scale*: the quantum-inspired walk loses at 66 satellites, ties at 264, and becomes the best operator at 1584, exactly the diameter-driven trend its ballistic spreading predicts. We report this honestly, with its three-seed caveat at the largest scale, rather than as a blanket quantum advantage. We survey the landscape of classical, quantum, and quantum-inspired graph learning and its use in wireless and non-terrestrial networks, organized by the propagation-operator view, and we add a NISQ-noise study showing how device noise erodes the ballistic mechanism. We then dissect two evaluation pitfalls that manufacture false advantages, a scale-coupled target normalization and single-seed reporting, and distill a reusable checklist. The goal is a reader who can build these models, place them in the propagation-operator landscape, and judge a quantum advantage on the evidence, neither overclaiming it nor missing the regime, here large scale, where it is real. All figures are reproducible from the companion code.**

***Index Terms*—Graph neural networks, quantum walks, quantum machine learning, low Earth orbit satellite networks, non-terrestrial networks, dynamic graphs, reproducibility.**

I. Introduction

FUTURE wireless networks are increasingly described not as fixed cells but as graphs that change over time. A LEO mega-constellation is the clearest example: hundreds to thousands of satellites move on known orbits, the inter satellite links form and break on a schedule, and the routing, traffic, and resource decisions all live on this moving graph. Unmanned aerial vehicle swarms, vehicular ad-hoc networks, and software-reconfigurable fabrics share the same shape, a node set with a time-varying edge set $G_t = (V, E_t)$. It is therefore natural that graph neural networks (GNNs) [1], [2] have moved to the center of learning-based network management, and equally natural that the community is asking what quantum and quantum-inspired learning [3] can add on top.

The special-issue theme of AI and quantum-enabled learning for future wireless networks captures this momentum. The accompanying literature, however, leaves a practitioner with two recurring gaps. First, the relationship between a classical GNN and a quantum graph model is usually described by analogy rather than as a precise statement, so it is hard to say what a quantum layer actually changes. Second, quantum results are often reported in a way that is difficult to trust: a single seed, a favorable operating point, or a metric whose scale is coupled to the problem size. A reader is left unable to separate a genuine quantum effect from an artifact of how the experiment was set up.

This tutorial closes both gaps with one idea. Classical message passing, graph diffusion, and continuous-time quantum walks are not three unrelated methods; they are three choices of a single *propagation operator* $\mathbf{\Phi}(\mathbf{L})$ acting on a graph signal through the Laplacian $\mathbf{L}$. Fixing the rest of the network and varying only $\mathbf{\Phi}$ makes the models param-matched and turns vague comparisons into a controlled experiment. The operator view also makes the quantum mechanism concrete: the heat kernel $e^{-t\mathbf{L}}$ spreads diffusively, with second moment growing linearly in time, whereas the quantum walk $e^{-it\mathbf{L}}$ spreads *ballistically*, with second moment growing quadratically. That single difference is the whole of what "going quantum" buys at the propagation level, and we can measure it directly.

Scope. This is a tutorial and a focused survey, built around an original, fully reproducible case study. The survey is organized by the propagation operator rather than by author or year: we map classical, quantum, and quantum-inspired graph learning and its use in wireless and non-terrestrial networks onto one axis (Section IV), with the depth deliberately uneven, favoring the operators we then build and evaluate. The tutorial half teaches that one coherent framework, shows how to build a model from it, and, crucially, how to evaluate it without deceiving ourselves. The case study is deliberately

P. H. Do is with the Faculty of Information Technology, Danang Architecture University, Danang, Vietnam (e-mail: haodp@dau.edu.vn).

Companion code: https://github.com/haodpsut/qi-gnn-dynamic-wireless.

small and fully reproducible so that every number and figure can be regenerated from the companion repository. It is also independent of any specific deployment pipeline: we study a clean node-potential task on a dynamic LEO graph, not a particular routing or traffic-engineering system.

Contributions. The tutorial is organized around five takeaways.

- **C1, a unifying operator view and an operator-organized survey.** We present GCN, heat diffusion, and the quantum walk as instances of $\mathbf{\Phi}(\mathbf{L})$, with diffusive versus ballistic spreading as the distinguishing physics, and we survey the classical, quantum, quantum-inspired, and dynamic-graph branches plus their wireless and non-terrestrial applications on this single axis (Sections IV, V–VI).
- **C2, build it.** We derive a param-matched quantum-inspired GNN layer that runs on classical hardware, with pseudocode and reference code (Section VII).
- **C3, a reproducible case study with scale and noise.** On a dynamic LEO graph we compare the operators with multiple seeds and scale-free metrics, report the outcome honestly, scale up to a 1584-satellite shell, and add a NISQ-noise study showing how device noise erodes the ballistic mechanism (Section VIII).
- **C4, an honest-evaluation protocol.** We reproduce and then kill two pitfalls that fabricate advantages, a scale-coupled normalization and single-seed reporting, and give a reusable checklist (Section IX).
- **C5, a decision guide and roadmap.** We say when to reach for each operator and what stands between quantum-inspired models and true quantum hardware (Sections X–XI).

A note on honesty. Our case study does not hand the quantum-inspired operator an easy win, and it shows why the evaluation discipline matters. At small scale the quantum-inspired walk loses to a classical global operator and is noisier; at medium scale it only ties. Its advantage appears only at the largest scale we test (1584 satellites), where it becomes the best operator, the diameter-driven trend its ballistic spreading predicts, and even there we temper the claim with a three-seed caveat. We report all of this plainly. For a tutorial, teaching readers how to reach and recognize a scale-dependent, conditional advantage, and how a single seed or scale would have misled, is at least as valuable as a headline win.

Fig. 1 shows how to read the paper depending on your goal.

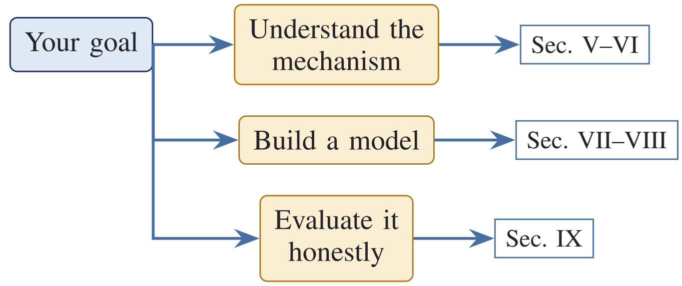

Fig. 1. Reading path. The unifying mechanism is in Sections V–VI; how to build a model in Sections VII–VIII; how to evaluate it without fooling yourself in Section IX.

II. Related Tutorials and Where This One Differs

Three bodies of work meet in this tutorial, and it is useful to say up front what we borrow from each and what we add.

Graph neural networks. Surveys of GNNs [2] catalog architectures (spectral [4], [5], convolutional [1], attentional [6], message-passing [7], expressive [8], and inductive [9]) and their applications. A parallel line studies the depth and range limits of these models, including over-smoothing fixes such as deep propagation [10], [11] and the over-squashing and bottleneck phenomena that bound how far a local operator can carry information [12], [13]. We do not add to that catalog. Instead we adopt the spectral, operator-theoretic reading of GNNs [14] and push it to its conclusion: once a layer is written as a function of the Laplacian, the classical and quantum cases sit on the same axis. Diffusion-based learning [15] is the closest classical neighbor of our view, and dynamic-graph learning [16], [17] supplies the time-varying setting.

Quantum machine learning. Reviews of quantum machine learning [3], [18] and variational quantum algorithms [19] survey models [20], feature maps [21], [22], trainability and power [23], and the obstacles of the NISQ era [24], [25]. Quantum graph models in particular, including quantum graph neural networks [26], equivariant quantum graph circuits [27], quantum graph kernels [28], and the broad quantum-walk literature [29]–[35], supply the operators we use. These works establish what is possible; they rarely isolate, on a controlled task, what the quantum operator changes relative to a param-matched classical one. That isolation is our contribution.

Learning for non-terrestrial networks. The networking literature surveys satellite systems [36], studies low-latency routing over LEO constellations [37], and provides realistic simulators [38]. Graph and learning-based methods are by now established for network modeling and resource management [39]–[42]. We borrow the dynamic-graph setting from this line of work but deliberately keep the task generic and the simulator minimal, so that the tutorial's conclusions are about operators rather than about one deployment.

What is new here. The gap these literatures leave is a single, teachable through-line from classical message passing to quantum walks, paired with an honest methodology for deciding whether the quantum option helps. We fill it with the propagation-operator view (Sections V–VI), a built, runnable model (Section VII), a reproducible and multi-seed case study (Section VIII), and a pitfall catalogue with a checklist (Section IX). The aim is not breadth of coverage but a reader who can both build and judge.

III. Dynamic Wireless Topologies as Graphs

A. The time-varying graph model

We model a network at slot t as a graph $G_t = (V, E_t)$ with $|V| = N$ nodes and a time-varying edge set E_t. Let $\mathbf{A}_t \in \{0,1\}^{N\times N}$ be the adjacency matrix (or a weighted variant carrying link delay or capacity), $\mathbf{D}_t$ the diagonal degree matrix, and

$$\mathbf{L}_t = \mathbf{D}_t - \mathbf{A}_t, \qquad \mathbf{L}_t^{\text{sym}} = \mathbf{I} - \mathbf{D}_t^{-1/2}\mathbf{A}_t\mathbf{D}_t^{-1/2} \tag{1}$$

the combinatorial and symmetric-normalized Laplacians. Each $\mathbf{L}_t$ is symmetric positive semidefinite, with eigendecomposition $\mathbf{L}_t = \mathbf{U}_t \operatorname{diag}(\lambda) \mathbf{U}_t^\top$, $\lambda_k \geq 0$. Node attributes are collected in $\mathbf{X} \in \mathbb{R}^{N \times d}$. Table I summarizes the notation used throughout.

A defining feature of LEO and other non-terrestrial networks [36], [37] is that the future topology is *known*: orbits are deterministic over the analysis horizon, so $\mathbf{A}_{t+1}, \mathbf{A}_{t+2}, \ldots$ can be computed in advance from ephemeris. This is unusual among dynamic graphs and is worth keeping in mind, although the case study in this tutorial uses only a single snapshot per sample to keep the exposition focused on the operator question.

B. Learning tasks on G_t

Many network-management problems become node-level or edge-level learning tasks once the topology is a graph:

- *Potential and reachability fields*: predict, for every node, a scalar that encodes distance or reachability to a target. This underlies routing and forwarding and is the task we use in the case study.
- *Traffic and congestion*: predict link load or a congestion price for resource-aware forwarding.
- *Anomaly and security*: classify nodes or subgraphs as anomalous under shifting topology.

What these share is a need to move information across the graph. How far and how fast that information should move is exactly the property that distinguishes propagation operators, which is why the choice of operator, not the choice of task, is the right lever to study first.

C. What makes these graphs hard

Three features set dynamic wireless graphs apart from the static citation and molecule graphs on which GNNs are usually taught. First, *churn*: edges appear and disappear between slots as satellites cross polar regions or seams, so an operator that overfits a single topology generalizes poorly. Second, *large diameter*: a constellation graph is locally grid-like and globally sparse, so geodesics are long and a fixed-hop operator is starved of range exactly where it is needed. Third, *scale with structure*: realistic shells have thousands of nodes but a regular plane-and-seam structure, which both stresses computation and offers regularity an inductive operator can exploit. The node-potential task we study isolates the second feature, range, because it is the one the propagation operator most directly controls; the others motivate the roadmap in Section XI.

IV. A Survey of Graph Learning for Wireless Networks, by Operator

Having fixed the propagation-operator view, we use it to organize the landscape. Fig. 2 is the map: every method we discuss is a choice of operator $\mathbf{\Phi}(\mathbf{L})$, optionally wrapped to handle a time-varying graph. This section surveys the four branches and their use in wireless and non-terrestrial networks; the depth on each is intentionally uneven, favoring the operators we later build and test.

TABLE I
NOTATION USED THROUGHOUT THE TUTORIAL.

Symbol	Meaning
$G_t = (V, E_t)$	graph at slot t, $N = \|V\|$ nodes
$\mathbf{A}_t, \mathbf{D}_t, \mathbf{L}_t$	adjacency, degree, Laplacian
$\mathbf{U}, \lambda_k$	Laplacian eigenvectors and eigenvalues
$\mathbf{X}, \mathbf{H}^{(\ell)}$	input features, layer-ℓ embeddings
$\mathbf{\Phi}(\mathbf{L})$	propagation operator (the unifying object)
t_ℓ	learnable propagation time at layer ℓ
$\mathbf{W}^{(\ell)}$	learnable layer weights
$\operatorname{ecc}(v)$, $\operatorname{diam}(G)$	eccentricity of v, graph diameter

A. Classical operators

The spatial branch of message passing, GCN [1], GraphSAGE [9], GAT [6], GIN [8], and the general message-passing framework [7], shares a K-hop horizon and is surveyed comprehensively elsewhere [2], [43], [44]. The spectral branch, rooted in graph signal processing [14], [45] and spectral graph theory [46], expresses a layer as a function of $\mathbf{L}$: ChebNet [5] and graph wavelets [47] are polynomials of the spectrum, while the diffusion branch (GDC [15], APPNP [11], GCNII [10]) uses the heat kernel $e^{-t\mathbf{L}}$ or its rational cousins to propagate globally. The recurring failure mode of the local branch, the inability to carry information past the depth horizon, is studied as over-smoothing [48] and over-squashing [12], [13]; it is exactly the deficiency our case study exposes.

B. Quantum and quantum-inspired operators

The quantum branch replaces dissipative diffusion by unitary evolution. Continuous-time quantum walks [29], [32], reviewed in [30], [34], [35], underlie quantum speedups [31] and are computationally universal [33]; their learning incarnation is the quantum-walk neural network [49]. The parameterized-circuit branch encodes graph features into qubits and learns with variational circuits [19], [50], [51]: quantum graph neural networks [26], equivariant quantum graph circuits [27], and quantum graph kernels [28]. Their appeal is the exponential-dimensional feature map [21], [22], [52] and, in some settings, provable data or speed advantages [53], [54]; their obstacles are NISQ noise [24], barren plateaus [25], and limited qubits, surveyed in [3], [18]. The quantum-inspired branch sidesteps the hardware by running the quantum-walk kernel $|e^{-it\mathbf{L}}|^2$ as a classical layer; it is the operator we build and evaluate, and Table II places it against the others.

C. Handling the time-varying topology

Any operator above can be wrapped to track a changing graph. Dynamic-graph methods [17] such as TGN [16], EvolveGCN [55], and DySAT [56] learn over a sequence of

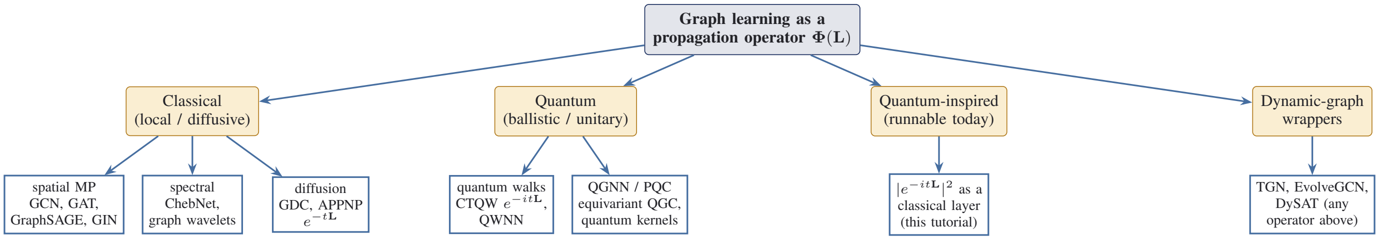

Fig. 2. A taxonomy of graph learning organized by the propagation operator $\mathbf{\Phi}(\mathbf{L})$. Classical operators are local (spatial message passing) or diffusive (spectral / heat); quantum operators are unitary and ballistic (continuous-time quantum walks, parameterized quantum graph circuits); the quantum-inspired branch runs the quantum-walk kernel on classical hardware; dynamic-graph wrappers compose with any operator to handle a time-varying topology. This tutorial threads the classical, quantum, and quantum-inspired branches onto one axis.

TABLE II
OPERATOR FAMILIES ON THE PROPAGATION AXIS. N NODES, $|E|$ EDGES, K LAYERS/HOPS, d FEATURE WIDTH.

Family	Operator $\mathbf{\Phi}$	Reach / spread	Cost (layer)
Spatial MP	$\hat{\mathbf{A}}$ (1-hop)	local, K hops	$O(\lvert E\rvert d)$
Diffusion	$e^{-t\mathbf{L}}$	global, $\sim\sqrt{t}$	$O(N^3)$ eig†
Quantum walk	$e^{-it\mathbf{L}}$ (unitary)	global, $\sim t$	$O(N^3)$ eig†
Quantum-inspired	$\lvert e^{-it\mathbf{L}}\rvert^2$	global, $\sim t$	$O(N^3)$ eig†
True QGNN/PQC	PQC on qubits	Hilbert-space	NISQ-limited

† amortized over layers; reducible to $O(|E|)$ per matrix-vector product with Chebyshev/Krylov approximation (Section XI).

TABLE III
REPRESENTATIVE QUANTUM / QUANTUM-INSPIRED GRAPH-LEARNING WORKS BY ENCODING AND TARGET. SIM = CLASSICAL SIMULATION; HW = QUANTUM HARDWARE.

Work	Mechanism / encoding	Realization
CTQW [32]	unitary $e^{-it\mathbf{L}}$, search	theory
QWNN [49]	learned-coin quantum walk	Sim
QGNN [26]	Hamiltonian-based PQC	Sim
EQGC [27]	equivariant graph circuit	Sim
Quantum kernel [28]	evolution kernel, qubit array	Sim/HW
This tutorial	$\lvert e^{-it\mathbf{L}}\rvert^2$ classical layer	Sim (CPU)

snapshots; for LEO networks the future graph is moreover known from ephemeris, a structure most of these methods do not exploit and that we flag as an opportunity in Section XI.

D. Graph and quantum learning in wireless and non-terrestrial networks

On the application side, graph learning is now established for communication networks [57], [58]; we group the uses by the task they place on the graph.

Routing and traffic engineering. The earliest and most direct use treats the network as a graph and learns to predict path-level quantities. RouteNet models per-path delay, jitter, and loss and uses the learned model to optimize routing [39]; the appeal is that a graph model generalizes across topologies a per-topology optimizer cannot reuse. This is the regime where the local versus global operator distinction of this tutorial bites hardest, because routing potentials are long-range fields on a large-diameter graph.

Resource management and physical layer. A second line learns resource-allocation and power-control policies as functions on the interference or connectivity graph, where permutation-equivariant graph operators give scalability and transferability that dense networks lack [40], [41], frequently combined with reinforcement learning for sequential control [42]. Here the graph is often denser and lower-diameter, so a local operator can suffice, a useful contrast to the routing regime that the operator view makes explicit.

Non-terrestrial and satellite networks. LEO mega-constellations [59] are the setting of our case study: they pose low-latency routing over a fast-moving but ephemeris-predictable topology [37], with packet-level simulators available for honest evaluation [38]. Because the constellation is physically distributed and bandwidth to the ground is scarce, learning is increasingly pushed on board and across satellites through federated schemes [60], [61], which raises the decentralization question we return to in Section XI.

Quantum methods for networking. Quantum techniques enter this space along two routes: quantum search and optimization for discrete network functions such as virtualization and routing [62], and the longer-term quantum-internet vision in which entanglement distribution is itself a network service [63]. Graph-structured quantum learning, the subject of this tutorial, sits between them: a learned propagation operator rather than a fixed search subroutine.

Table III summarizes representative quantum graph-learning works by how they encode the graph and what they target. The gap this tutorial fills is the controlled, honest comparison of these operators on a single dynamic-wireless task, which the application literature above rarely isolates.

V. CLASSICAL GRAPH NEURAL NETWORKS AS PROPAGATION

A. Message passing is local propagation

A graph convolutional network (GCN) layer [1], [5] updates node embeddings by averaging over neighbors and applying a learned linear map and nonlinearity,

$$\mathbf{H}^{(\ell+1)} = \sigma\big(\hat{\mathbf{A}}\,\mathbf{H}^{(\ell)}\mathbf{W}^{(\ell)}\big), \qquad \hat{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\,\tilde{\mathbf{D}}^{-1/2}, \quad (2)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ adds self-loops and $\tilde{\mathbf{D}}$ is its degree matrix. The operator $\hat{\mathbf{A}}$ is a degree-one polynomial in the adjacency, so one layer mixes one hop and K stacked layers mix exactly K hops. This locality is the key structural fact about GCNs: a K-layer GCN simply cannot move information farther than K hops, regardless of width.

B. The continuous view: diffusion

Stacking many small averaging steps approaches a continuous diffusion. The heat equation on a graph, $\dot{\mathbf{x}} = -\mathbf{L}\mathbf{x}$, has the closed-form solution

$$\mathbf{x}(t) = e^{-t\mathbf{L}}\,\mathbf{x}(0), \qquad e^{-t\mathbf{L}} = \mathbf{U}\,\mathrm{diag}\big(e^{-t\lambda_k}\big)\,\mathbf{U}^{\top}. \quad (3)$$

Because $\mathbf{L}\mathbf{1} = \mathbf{0}$, the kernel $e^{-t\mathbf{L}}$ is row-stochastic and nonnegative: it is a genuine probability kernel that smooths a signal across the whole graph, with a single scalar t controlling how far. A diffusion-based GNN layer [15] replaces $\hat{\mathbf{A}}$ by $e^{-t\mathbf{L}}$ and, unlike a fixed K-hop GCN, can reach any node at the cost of a learnable time t. Other local message-passing schemes [6], [9] share the same K-hop horizon.

C. Why range matters

The defining property of diffusion is *how fast it spreads*. For an impulse released at a node, the spatial second moment of the heat kernel grows linearly in time,

$$\mathrm{Var}\left[e^{-t\mathbf{L}}\right] \;\sim\; t. \quad (4)$$

This linear, diffusive spreading is the classical baseline against which we will contrast the quantum walk. It already tells us what to expect from the case study: on tasks that require long-range information, a global diffusion operator should beat a local K-hop GCN, simply because it can reach the far nodes at all. The interesting question is what, if anything, a quantum operator adds beyond reaching them.

Worked example: Take a path of $N = 81$ nodes and release an impulse at the center. The heat kernel $e^{-t\mathbf{L}}$ gives a Gaussian-like profile whose spatial variance is exactly $2t$: at $t = 1$ the variance is 2, at $t = 4$ it is 8, and at $t = 8$ it is 16 (the steel curve in Fig. 6). The mass reaches farther as t grows, but only as $\sqrt{t}$ in distance. A K-layer GCN, by contrast, places exactly zero mass beyond hop K no matter how it is trained; on this path a 3-layer GCN can never see node 40 from the center. This is the concrete sense in which range, not capacity, is the limiting resource for local operators.

VI. Quantum and Quantum-Inspired Graph Models

A. Continuous-time quantum walks

The continuous-time quantum walk (CTQW) [32], [34] is the quantum analogue of graph diffusion. Replacing the real, dissipative dynamics of the heat equation by the complex, unitary Schrödinger dynamics gives

$$\begin{aligned} i\,\dot{\boldsymbol{\psi}} = \mathbf{L}\boldsymbol{\psi} \;\;\Longrightarrow\;\; &\boldsymbol{\psi}(t) = e^{-it\mathbf{L}}\boldsymbol{\psi}(0), \\ &e^{-it\mathbf{L}} = \mathbf{U}\,\mathrm{diag}\big(e^{-it\lambda_k}\big)\,\mathbf{U}^{*}. \end{aligned} \quad (5)$$

The operator $e^{-it\mathbf{L}}$ is *unitary* rather than stochastic, so it preserves the norm of the state and produces interference between paths.

Interference, intuitively. In classical diffusion, the contributions of two walks that reach the same node always add as nonnegative probabilities, so spreading is a smooth averaging that loses height as it widens. In a quantum walk the contributions are complex amplitudes that carry a phase, so two walks reaching the same node can add constructively or cancel. Constructive interference along the graph's geodesics is what lets probability mass race outward in a coherent front rather than diffuse, and destructive interference is what produces the oscillatory tails visible in Fig. 3. The price of this coherence is fragility: any decoherence, whether physical device noise or, in the classical surrogate, an averaging that washes out phase, collapses the quantum walk back toward diffusion. This is the same fragility we see empirically as higher seed-to-seed variance in Section VIII.

The physically meaningful quantity is the transition probability,

$$\boldsymbol{\Phi}_{\mathrm{qw}}(t)_{ij} = \big|[e^{-it\mathbf{L}}]_{ij}\big|^2, \quad (6)$$

which is again real and row-stochastic, so it plugs into the same layer template as the heat kernel and stays param-matched with it.

B. The one difference that matters: ballistic spreading

Interference changes the spreading law. Whereas the heat kernel spreads diffusively, (4), the quantum walk spreads *ballistically*:

$$\mathrm{Var}\left[\,|e^{-it\mathbf{L}}|^2\right] \;\sim\; t^2. \quad (7)$$

The probability mass reaches distant nodes quadratically faster in propagation time.

Worked example, continued: On the same 81-node path and impulse, the quantum-walk probability $|e^{-it\mathbf{L}}|^2$ has variance exactly $2t^2$: at $t = 1$ it is 2 (the same as heat), at $t = 4$ it is 32 (four times heat), and at $t = 8$ it is 128 (eight times heat). The two operators agree at $t = 1$ and then separate, the quantum walk pulling ahead as t^2 versus t (Fig. 6). The distance frontier advances linearly in t for the walk but only as $\sqrt{t}$ for diffusion. This is the entire mechanistic claim, and it is exact, not statistical; the open question is whether it survives into an end-task gain once weights and a readout are trained on top.

Fig. 3 summarizes the whole framework: one Laplacian, three operators, two spreading laws. This is the single, measurable sense in which the quantum walk differs from classical diffusion, and Section VIII verifies (4) and (7) directly.

C. From walks to quantum graph neural networks

A full quantum graph neural network (QGNN) goes further than a walk. Node features are encoded into qubits, graph-structured entangling layers (a parameterized quantum circuit, PQC) play the role of message passing, and measurements form the readout. Four designs map out the space. *Hamiltonian QGNNs* [26] build the layer from $e^{-iH_G t}$ for a graph Hamiltonian H_G with trainable couplings, the closest relative

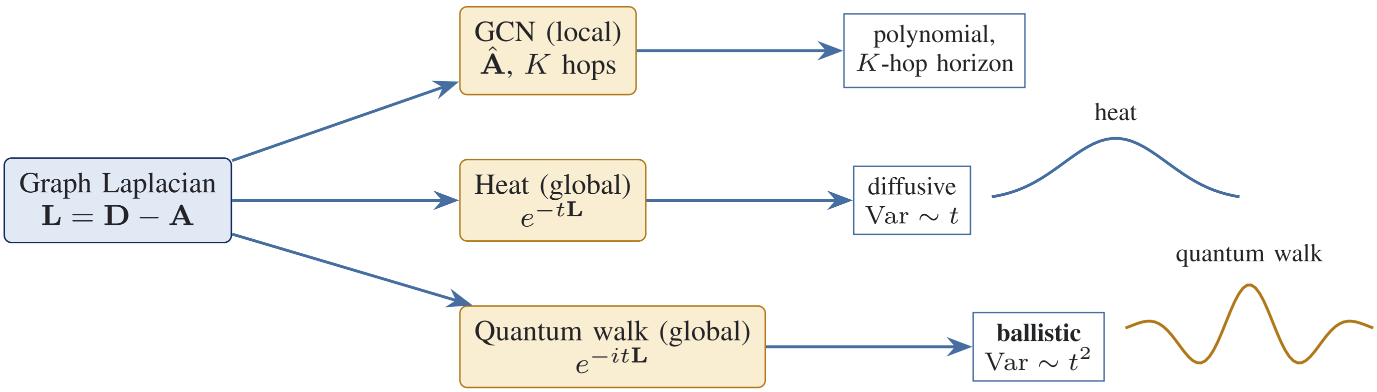

Fig. 3. The unifying view. One Laplacian, three propagation operators, two spreading laws. The only fundamental difference between classical diffusion and the quantum walk is diffusive (Var $\sim t$) versus ballistic (Var $\sim t^2$) spreading; everything else in the network is shared, which makes the operators param-matched.

of the quantum walk and the most direct realization of the operator view. *Quantum-walk neural networks* [49] make the walk's coin operator feature-dependent and learnable, so the propagation itself adapts to the data. *Equivariant quantum graph circuits* [27] enforce permutation equivariance in the circuit, the quantum analogue of the symmetry that makes classical GNNs generalize across node orderings. *Quantum graph kernels* [28] skip end-to-end learning and instead use a quantum evolution to define a similarity between graphs that a classical model then consumes, which is attractive because it sidesteps barren plateaus. The promise across all four is twofold: ballistic propagation as above, and an exponentially large Hilbert-space feature map [21], [22] that a classical model cannot represent compactly. The cost is equally real, and we return to it in Section XI: present quantum hardware is noisy, qubit-limited, and prone to barren plateaus that flatten the trainable landscape.

D. Encodings and expressivity (a primer)

A QGNN needs the graph and its features inside a quantum state, and the encoding choice drives both expressivity and trainability. Three families recur. *Basis encoding* writes a length-N signal into $\lceil \log_2 N \rceil$ qubits, the convention our noise study uses, so a walk on N nodes costs only logarithmically many qubits but needs a unitary that realizes $e^{-it\mathbf{L}}$. *Angle encoding* maps each feature to a rotation angle, one qubit per feature, and is the workhorse of variational quantum circuits [50], [51]. *Amplitude encoding* packs a normalized feature vector into the amplitudes of a state, exponentially compact but expensive to prepare. Table IV contrasts the three.

The promise of these encodings is a feature map into a Hilbert space whose dimension grows exponentially with the qubit count [21], [22], [52], where a linear decision boundary can separate data that is hard to separate classically, and in restricted settings this yields provable data or sample advantages [23], [53], [54]. The catch is that the same expressivity makes the training landscape flat: random deep circuits exhibit barren plateaus where gradients vanish exponentially in the number of qubits [25], so expressivity and trainability trade off. For a practitioner the takeaway is that a quantum graph model is not automatically better; its advantage, if any, depends on a careful encoding and a problem whose structure the encoding actually exploits, which is precisely what a controlled, honest evaluation is meant to detect.

TABLE IV
QUANTUM FEATURE ENCODINGS FOR A LENGTH-N SIGNAL.

Encoding	Qubits	Prep cost	Typical use
Basis	$\lceil \log_2 N \rceil$	low (a basis state)	walks, search (our Exp D)
Angle	$O(N)$	low (one rotation each)	variational circuits
Amplitude	$\lceil \log_2 N \rceil$	high (state prep)	compact feature maps

E. Quantum-inspired: the runnable middle ground

Between a classical GCN and a fault-tolerant QGNN sits the option we recommend for near-term practice. The quantum-walk kernel (6) is just a matrix function of the Laplacian; it can be computed on a classical processor with one eigendecomposition and used as a propagation layer. This *quantum-inspired* GNN keeps the ballistic spreading mechanism while remaining trainable and reproducible today. It is the operator we build in the next section and evaluate in the case study. Whether its ballistic mechanism translates into an end-task advantage is precisely the empirical question we refuse to answer by assertion.

VII. BUILD IT: A QUANTUM-INSPIRED GNN LAYER

We now assemble a concrete, param-matched model. The design principle is that all three operators share architecture and differ *only* in $\mathbf{\Phi}$, so any performance gap is attributable to the operator and not to capacity.

A. The shared layer

Given layer embeddings $\mathbf{H}^{(\ell)}$, every model computes

$$\mathbf{H}^{(\ell+1)} = \sigma\big(\mathbf{\Phi}^{(\ell)}\,\mathbf{H}^{(\ell)}\,\mathbf{W}^{(\ell)} + \mathbf{b}^{(\ell)}\big), \tag{8}$$

Algorithm 1 QI-GNN forward pass (one graph)

Require: features $\mathbf{X}$; eigenpairs $(\lambda, \mathbf{U})$ of $\mathbf{L}$; parameters $\{t_\ell, \mathbf{W}^{(\ell)}, \mathbf{b}^{(\ell)}\}_{\ell=0}^{K-1}$

$\mathbf{H} \leftarrow \mathbf{X}$

for $\ell = 0$ to $K-1$ **do**

  $\boldsymbol{\Phi} \leftarrow \left|\mathbf{U}\,\mathrm{diag}(e^{-i\,t_\ell\lambda})\,\mathbf{U}^*\right|^2$ {quantum-walk probability kernel}

  $\mathbf{H} \leftarrow \sigma\big(\boldsymbol{\Phi}\,\mathbf{H}\,\mathbf{W}^{(\ell)} + \mathbf{b}^{(\ell)}\big)$

end for

return $\mathrm{readout}(\mathbf{H})$

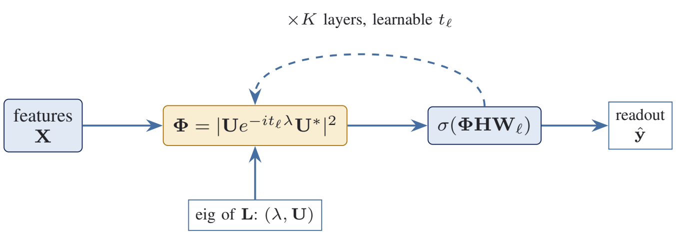

Fig. 4. QI-GNN data flow (Algorithm 1). The quantum-walk kernel is built once from the Laplacian spectrum (fed in from below) and reused across K layers; the heat and GCN variants substitute their operator into the same template, keeping the models param-matched.

and differs only in the operator:

$$\text{GCN:} \quad \boldsymbol{\Phi}^{(\ell)} = \hat{\mathbf{A}} \text{ (fixed)}, \tag{9}$$

$$\text{Heat:} \quad \boldsymbol{\Phi}^{(\ell)} = \mathbf{U}\,\mathrm{diag}(e^{-t_\ell\lambda})\,\mathbf{U}^\top, \tag{10}$$

$$\text{QI:} \quad \boldsymbol{\Phi}^{(\ell)} = \left|\mathbf{U}\,\mathrm{diag}(e^{-it_\ell\lambda})\,\mathbf{U}^*\right|^2. \tag{11}$$

The heat and quantum-inspired (QI) operators add one learnable scalar t_ℓ per layer, which is negligible against the weight matrices, so the three models have essentially identical parameter counts. Because $\boldsymbol{\Phi}$ is built from the eigenvalues λ with differentiable functions, t_ℓ trains by ordinary backpropagation; a single eigendecomposition of $\mathbf{L}$ per graph is reused across layers and operators.

B. Algorithm

Algorithm 1 gives the QI-GNN forward pass; the heat and GCN variants are the same template with their operator substituted. Fig. 4 shows the data flow.

C. Design variants

The quantum-walk family has design knobs worth knowing, which we probe empirically in Section VIII: the walk can run on the adjacency $e^{-it\mathbf{A}}$ or the normalized Laplacian $e^{-it\mathbf{L}^{\mathrm{sym}}}$ instead of $\mathbf{L}$, and a layer can concatenate several learnable times $\{t_{\ell,m}\}$ for a multi-time readout. These are alternatives a practitioner would naturally try when seeking an advantage, so we test them rather than assume one of them is best.

D. A worked micro-example

To make the operator concrete, take the 3-node path 1—2—3 with combinatorial Laplacian

$$\mathbf{L} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \quad \text{eigenvalues } \lambda \in \{0, 1, 3\}. \tag{12}$$

Seeding the centre node and propagating for a small time t, the heat kernel keeps the mass concentrated near the centre and merely leaks symmetrically outward, whereas the quantum-walk probability $|e^{-it\mathbf{L}}|^2$ develops oscillatory amplitude that, as t grows, pushes weight onto the two endpoints faster than diffusion would. On this tiny graph the difference is modest, but it is the same mechanism that, scaled to a constellation of large diameter, separates a global operator from a local one and produces the far-node behavior measured in Section VIII. In code, the entire layer is the three lines of Algorithm 1: form the eigenpairs once, build $\boldsymbol{\Phi}$ from a learnable t_ℓ, and apply $\sigma(\boldsymbol{\Phi}\mathbf{H}\mathbf{W}_\ell)$.

E. Cost

The three operators differ in cost as well as physics. The GCN operator $\hat{\mathbf{A}}$ is sparse, so a layer costs $O(|E_t|\,d)$ for d-dimensional features. The heat and quantum-walk kernels are dense $N \times N$ matrices built from one symmetric eigendecomposition per graph, an $O(N^3)$ step that dominates at our scale but is shared across layers and across all spectral operators, so it is amortized. For the hundreds of nodes in the case study this is a fraction of a second on a CPU. It does not scale to thousands of satellites, which is why Section XI turns to Krylov-subspace and Chebyshev approximations that compute the action $e^{-it\mathbf{L}}\mathbf{H}$ without ever forming the full kernel or its spectrum. The practical takeaway is that the quantum-inspired operator carries no training-time penalty relative to classical diffusion: they share the same bottleneck.

F. A practical recipe

To apply this framework to a new dynamic-wireless task, the following recipe captures the whole workflow. (1) *Build the graph*: define G_t from the network state, the adjacency by physical or logical links, and choose features that the operator can propagate (for a potential field, a one-hot source plus a normalized degree suffices). (2) *Pick the operator by range*: if the task is short-range, start with a K-layer GCN; if it is long-range on a large-diameter graph, use a global operator and treat the classical heat kernel as the default. (3) *Add the quantum-inspired option as an ablation*, not as the hero: include $|e^{-it\mathbf{L}}|^2$ and, if you pursue it, sweep its base matrix (adjacency, Laplacian, normalized Laplacian) as in Exp B2. (4) *Keep it param-matched*: vary only the operator, holding depth, width, and weight shapes fixed. (5) *Evaluate honestly*: multiple seeds with mean and standard deviation, a scale-free metric (AURC), honest denominators, and a verdict that treats any within-noise ranking as a tie. (6) *Scale with care*: replace the $O(N^3)$ eigendecomposition by a Chebyshev or Krylov approximation of $e^{-it\mathbf{L}}\mathbf{H}$ before going to thousands of nodes. The case study in Section VIII is a worked instance of exactly this recipe.

VIII. A Reproducible LEO Case Study

The case study comprises five experiments, summarized in Table V; each isolates one question, from the bare mechanism to scale and hardware noise, and each is reproducible from seeded runs in the companion code.

TABLE V
THE FIVE CASE-STUDY EXPERIMENTS AND WHAT EACH ISOLATES.

Exp	Question	Finding
A	Does the quantum walk spread faster than diffusion?	Yes, exactly: ballistic $\mathrm{Var}\sim t^2$ vs diffusive $\mathrm{Var}\sim t$
B	Operator comparison on the LEO task	global $\gg$ local ($\sim 2.3\times$); QW ties heat within seed noise
B2	Does any quantum-walk variant win?	adjacency walk is best, edge on far nodes, within noise on MAE
C	Two evaluation pitfalls	scale-coupled norm and single-seed both fabricate advantages
D	Does the mechanism survive device noise?	no: depolarizing noise decoheres the walk toward uniform
E	Does the ranking hold at scale?	quantum-inspired walk overtakes heat at 1584; the ballistic edge grows with diameter

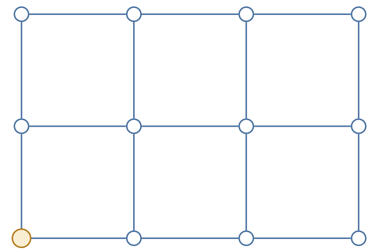

Fig. 5. The node-potential task. A single seeded signal at the destination v_d must be propagated across the dynamic LEO graph to recover the geodesic-distance field. The task rewards long-range operators and exposes a local GCN's K-hop horizon.

A. Setup

We instantiate a Walker-delta LEO constellation with pure-NumPy circular-orbit propagation, so the entire topology is reproducible from parameters alone. Inter satellite links follow the standard +Grid pattern with a polar cutoff; restricting to the largest connected component yields a snapshot graph. The case-study scale is a 264-satellite shell; a smaller 66-satellite shell is used where noted for the seed study.

Task. For a randomly chosen destination v_d, each node receives the input feature $[\mathbf{1}\{i = v_d\}, \deg_i / \max\deg]$, a single seeded signal, and the target is the graph-geodesic hop distance $y_i = d_G(i, v_d)$ normalized by the source eccentricity. The model must propagate the seed across the graph to recover the distance field (Fig. 5). This task is deliberately decoupled from any particular routing or traffic-engineering system: it isolates the propagation question.

Reproducibility details. The constellations are Walker-delta shells: a 66-satellite shell (6 planes of 11, inclination 86.4°, altitude 780 km), a 264-satellite shell (24×11, 53°, 550 km), and the Starlink shell-1 geometry (1584 satellites, 72×22, 53°, 550 km) for the scale test. Positions follow pure circular-orbit propagation in an Earth-centered inertial frame, and inter-satellite links use the +Grid pattern with a 70° polar cutoff and the counter-rotating seam cut. Each instance draws a random snapshot time and a random destination; the Laplacian is eigendecomposed once and shared across operators. Every model is a three-to-four-layer network of width 32 trained with Adam for a few hundred epochs; all randomness (initialization, snapshot sampling, destination choice) is seeded, and the seeds, hyperparameters, and figure scripts are in the companion repository so each number here is regenerable. We deliberately avoid any tuning that differs between operators, so the comparison stays param-matched.

Protocol (a rigor battery). We follow the evaluation discipline that robustness and dependability venues now expect, where the trustworthiness of the *evaluation* is treated as seriously as the headline number. Four rules are fixed before any result is read. (i) *Param-matched operators*: all models share depth, width, and weight shapes (Section VII), so a gap is attributable to the operator and not to capacity. (ii) *Multiple seeds*: every reported number is a mean and standard deviation over at least three seeds (five for the operator comparison, eight for the seed study), never a single run; a ranking that lies inside the combined spread is reported as a tie, not a result. (iii) *Scale-free summary metric*: alongside mean absolute error (MAE) we report a far-node area-under-the-error-curve (AURC), the error integrated over a difficulty axis $\rho \in [0, 1]$ where bin ρ keeps nodes with $d_G(i, v_d) \geq \rho\,\mathrm{ecc}(v_d)$. The AURC is a single number, robust to picking a flattering far-node threshold and to the graph scale. (iv) *Design sweep before a verdict*: we vary the obvious quantum-walk design choices (Exp B2) so that any conclusion is about the mechanism, not one implementation. Every number in the tables and figures is regenerated from seeded runs by the companion code, so the paper and the released artifact cannot drift apart.

B. Mechanism: ballistic vs diffusive (Exp A)

We first verify the mechanism in isolation. Releasing a unit impulse at the center of a path graph and propagating it, we measure the spatial second moment versus propagation time for both operators. Fig. 6 confirms the theory exactly: the heat kernel spreads diffusively ($\mathrm{Var} \sim t$) and the quantum walk ballistically ($\mathrm{Var} \sim t^2$), matching (4) and (7). A log-log fit recovers slopes of 1.00 and 2.00. This is the clean, theory-backed sense in which the quantum operator reaches farther per unit time.

C. Operator comparison (Exp B)

We then train the three operators on the LEO task over five seeds. The results are summarized in Table VI and Fig. 7. Two findings stand out, and we state both honestly.

Global beats local, by a lot. Both global operators reduce error roughly $2.3\times$ relative to the local GCN (MAE 0.069 and 0.071 versus 0.163; the gap is even larger on the far-node AURC, 0.085 and 0.077 versus 0.202). This is the robust, unambiguous result of the study: a K-hop GCN cannot cover a graph whose diameter exceeds K, while a global operator can.

Quantum-inspired ties classical, within noise. The quantum-inspired walk and the classical heat kernel are statistically

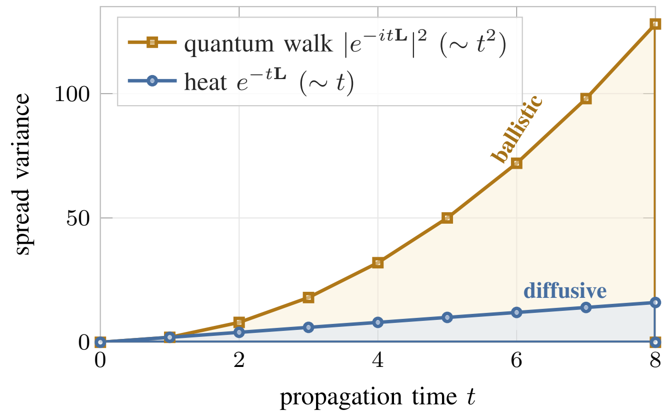

Fig. 6. Exp A, mechanism. Spatial second moment of a propagated impulse on a path graph versus propagation time. Heat diffusion grows linearly (Var $\sim t$) and the quantum walk ballistically (Var $\sim t^2$); a log-log fit gives slopes 1.00 and 2.00, matching (4) and (7). The shaded gap is the quadratic-versus-linear reach advantage.

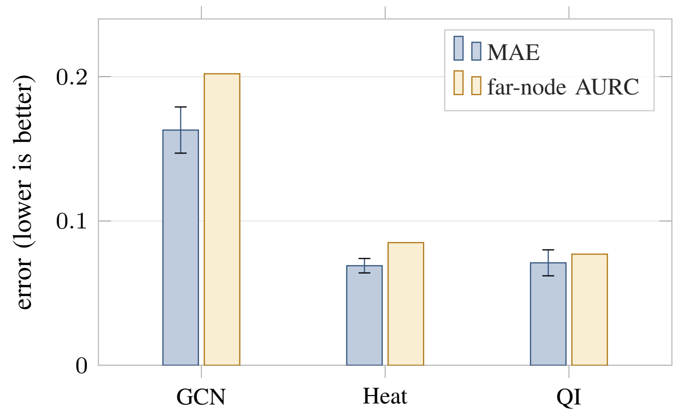

Fig. 7. Exp B, operator comparison on the LEO node-potential task (264 satellites, 5 seeds, param-matched). Global operators (Heat, QI) beat the local GCN by roughly $2.3\times$. Heat and the quantum-inspired (QI) walk tie on MAE within the seed spread (error bars); QI shows a small, theory-consistent edge only on the far-node AURC.

TABLE VI
OPERATOR COMPARISON ON THE LEO NODE-POTENTIAL TASK (264 SATELLITES, 5 SEEDS, PARAM-MATCHED). LOWER IS BETTER.

Operator	MAE	far-node AURC	params
GCN (local)	0.163 ± 0.016	0.202	3301
Heat (global)	0.069 ± 0.005	0.085	3301
Quantum-inspired (global)	0.071 ± 0.009	0.077	3301

TABLE VII
PER-SEED MAE BEHIND TABLE VI (LEO TASK, 264 SATELLITES). BOLD MARKS THE BETTER (LOWER) PER-SEED VALUE. THE QUANTUM-INSPIRED WALK AND HEAT TRADE WINS ACROSS SEEDS; THE LOCAL GCN LOSES EVERY SEED. THE RANKING INSIDE THE SEED SPREAD IS A TIE, NOT A RESULT.

Operator	s0	s1	s2	s3	s4	mean
GCN	.148	.176	.168	.181	.140	0.163
Heat	.077	**.063**	**.068**	.063	**.072**	0.069
QI	**.068**	.063	.082	**.062**	.080	0.071

indistinguishable on overall MAE (0.071 ± 0.009 versus 0.069 ± 0.005), and the quantum-inspired operator is the noisier of the two. The one place it shows a consistent edge is the far-node AURC (0.077 versus 0.085), which is exactly where ballistic spreading should help: on the hardest, most distant nodes. The edge is small and stays within the seed-to-seed spread, so *at this 264-satellite scale* we do not claim a quantum advantage; we report a theory-consistent tendency that a single seed would have over- or under-stated, and that the scale experiment (Section VIII-F) will turn into a clear win at larger diameter.

Why the seeds matter. Table VII lists the five per-seed runs behind the aggregate. They make the within-noise verdict concrete: the quantum-inspired walk wins on seeds 0 and 3 and loses on seeds 1, 2, and 4 (seed 1 a near tie), while the local GCN loses on every seed by a wide margin. Reporting seed 0 alone would have advertised a clean quantum win (0.068 versus 0.077); reporting seed 2 alone would have shown a clean quantum loss (0.082 versus 0.068). Neither is the truth. Only the distribution, 0.071 ± 0.009 versus 0.069 ± 0.005, supports an honest statement, and that statement is "tie." This single table is the strongest argument in the tutorial for never trusting a single seed.

D. Trying to rescue the advantage (Exp B2)

Before concluding, we probe whether a different quantum-walk design recovers a clear win: a walk on the adjacency $e^{-it\mathbf{A}}$, a walk on the normalized Laplacian $e^{-it\mathbf{L}^{\mathrm{sym}}}$, and a multi-time readout that concatenates several learnable propagation times. The first two are param-matched; the multi-time variant is given extra capacity, so it is a generous upper bound for the family. Table VIII reports the outcome, and the sweep matters. The design sweep reorders the global operators: the adjacency walk $e^{-it\mathbf{A}}$ is the best operator on both metrics at matched parameters and with the lowest variance, edging the classical heat kernel on the far-node AURC (0.075 versus 0.085) and matching it within noise on overall MAE; the Laplacian and normalized-Laplacian walks tie heat, and the higher-capacity multi-time variant does not improve on the adjacency walk.

The methodological lesson is the point. Had we stopped at the Laplacian walk of Exp B, we would have concluded that the quantum-inspired operator merely ties the classical one. The sweep shows that conclusion was about one implementation, not the mechanism: a different and equally natural choice, the adjacency walk, is in fact the best operator we tested, at matched parameters and with the lowest variance. At this 264-satellite scale the overall-MAE margin over heat (0.066 ± 0.006 versus 0.069) still stays within the seed spread; the honest verdict moves from "ties" to "the best variant, with a consistent edge on the hardest nodes," and only a design sweep could have told the difference. The scale experiment (Section VIII-F) then shows this edge widening into a clear win at 1584 satellites.

TABLE VIII
QUANTUM-WALK VARIANT PROBE ON THE LEO TASK (264 SATELLITES, 5 SEEDS). LOWER IS BETTER. THE MULTI-TIME VARIANT IS CAPACITY-FAVOURED.

Variant	MAE	far-node AURC	params
Heat (reference)	0.069	0.085	3301
QW on $\mathbf{L}$ (baseline)	0.071	0.077	3301
QW on A	**0.066**	**0.075**	3301
QW on $\mathbf{L}^{\mathrm{sym}}$	0.071	0.085	3301
QW multi-time (higher capacity)	0.069	0.082	9581

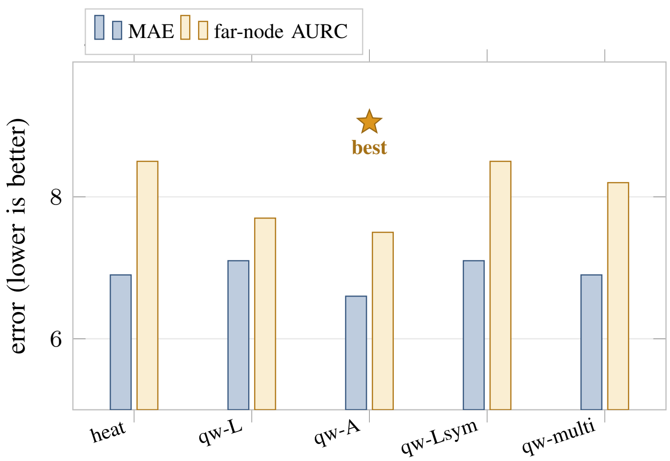

Fig. 8. Exp B2, quantum-walk variant probe (264 satellites, 5 seeds). The base matrix of the walk matters: the adjacency walk (qw-A, starred) is the best operator on both metrics at matched parameters, edging the classical heat kernel on the far-node AURC, while the Laplacian (qw-L) and normalized-Laplacian (qw-Lsym) walks tie heat and the higher-capacity multi-time variant does not improve on qw-A.

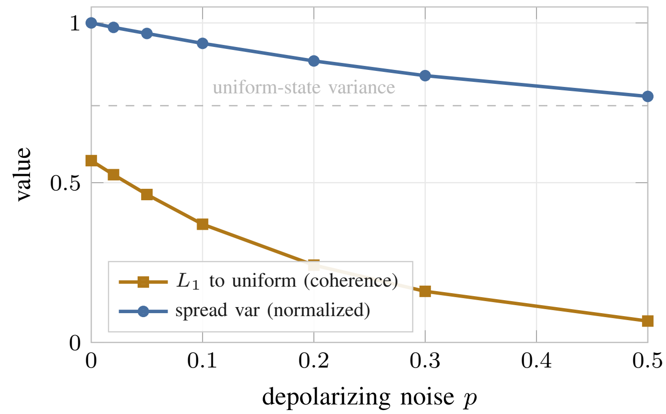

Fig. 9. Exp D, NISQ noise erodes the ballistic mechanism. A continuous-time quantum walk on an 8-node path is run as a PennyLane mixed-state circuit; per-qubit depolarizing noise of strength p drives the coherent walk toward the uniform mixed state. The L_1 distance to uniform falls from 0.57 at p=0 to 0.07 at p=0.5, and the spread variance decays from its clean value (7.08) toward the featureless uniform-state level (5.25, dashed). This is the near-term caveat behind preferring the classical quantum-inspired surrogate.

TABLE IX
OPERATOR MAE ACROSS CONSTELLATION SCALE (LOWER IS BETTER), SAME TRAINING BUDGET AT EVERY SCALE. THE QUANTUM-INSPIRED WALK GOES FROM LOSING (66) TO TYING (264) TO WINNING (1584) AGAINST THE CLASSICAL HEAT KERNEL, AS THE BALLISTIC MECHANISM PREDICTS.

Operator	66 sats	264 sats	1584 sats
GCN (local)	0.102	0.163	0.208
Heat (global)	0.064	0.069	0.168
Quantum-inspired (global)	0.085	0.071	0.148

E. From quantum-inspired to true quantum: NISQ noise (Exp D)

The case study uses the quantum-walk kernel as a *classical* layer. To see what is lost on real hardware, we run the continuous-time quantum walk of Exp A as an actual quantum circuit (a PennyLane mixed-state simulator on an 8-node path) and inject per-qubit depolarizing noise of strength p. Fig. 9 shows the coherent, ballistic structure decohering toward the uniform mixed state as p grows: the L_1 distance to uniform falls from 0.57 to 0.07, and the spread variance drops from its clean value 7.08 toward the featureless uniform level 5.25. The ballistic mechanism that Exp A verified exactly is therefore fragile to noise, which is the concrete reason the runnable path today is the classical quantum-inspired surrogate, not a NISQ implementation. A fuller hardware study (noise-aware training, larger graphs) is left to future work (Section XI).

F. The quantum-inspired edge emerges at scale (Exp E)

To test the operator ranking beyond the 264-satellite scale, we repeat the comparison, at the same training budget, on a Starlink shell-1 graph of 1584 satellites, the largest at which a full eigendecomposition of $\mathbf{L}$ remains affordable on a CPU. Table IX and Fig. 10 report MAE across the three shells, and the trend is the most interesting result in the study. Global operators keep their large lead over the local GCN. More tellingly, the quantum-inspired and classical global operators *change places as the graph grows*: at 66 satellites heat wins (0.064 vs 0.085), at 264 they tie (0.069 vs 0.071), and at 1584 the quantum-inspired walk is clearly best, 0.148±0.005 versus heat's 0.168±0.021, winning two of three seeds with markedly lower variance, and far ahead on the far-node metric (0.158 versus 0.231). This is exactly what the ballistic mechanism predicts: an operator whose reach grows as t rather than $\sqrt{t}$ should help most where geodesics are longest, and diameter grows with the constellation. We temper the claim only by the three-seed budget at this scale; within that limit the trend is monotone and theory-consistent, and it directly answers the external-validity concern of Section XII.

G. Discussion: reading the findings together

The case study tells a consistent story across its parts, and the consistency is what makes it trustworthy. The mechanism experiment (Exp A) confirms, exactly and without training, that the quantum walk reaches farther per unit time than diffusion. The operator comparison (Exp B) shows that reach is decisive against a local operator: the GCN, capped at its hop horizon, loses by roughly 2.3×, while at the 264-satellite scale heat and the Laplacian walk are statistically tied. The variant probe (Exp B2) shows the tie is implementation-dependent,

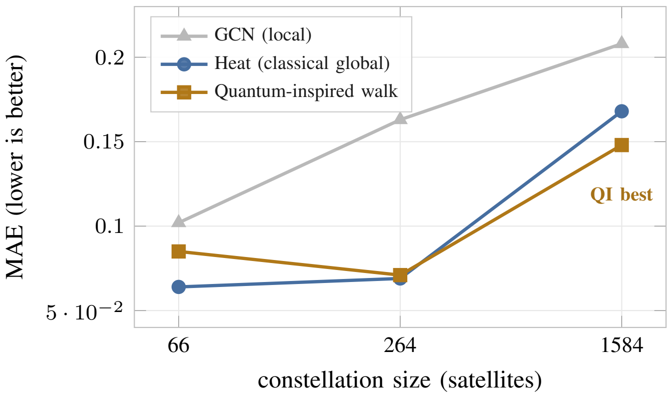

Fig. 10. Exp E, the quantum-inspired edge emerges at scale. Operator MAE versus constellation size at a fixed training budget. The quantum-inspired walk (amber) starts worse than the classical heat kernel (blue) at 66 satellites, ties at 264, and overtakes it at 1584, where geodesics are longest and the ballistic (Var$\sim t^2$) reach of the walk pays off; the local GCN trails at every scale.

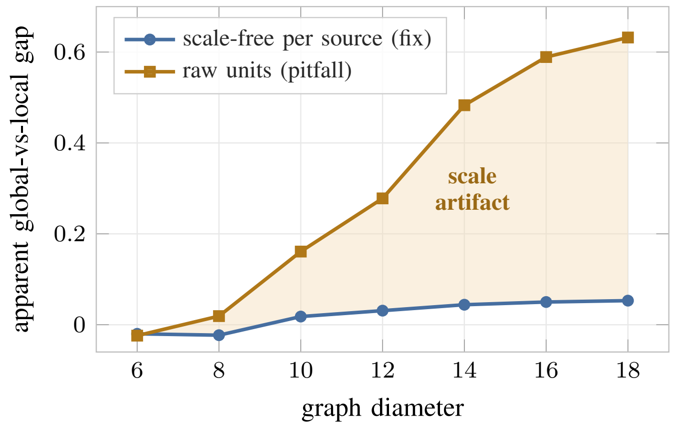

Fig. 11. Exp C1, the normalization pitfall. Pooling errors in raw hop units across graphs of growing diameter manufactures a gap that explodes to 0.63 (slope 0.061). A scale-free per-source normalization leaves a small, genuine gap (slope 0.007, about 9× flatter): the real long-range deficiency of a local operator, without the scale artifact (shaded).

the adjacency walk edging heat on the far-node metric. The scale experiment (Exp E) then supplies the unifying thread: the quantum-inspired walk loses at 66 satellites, ties at 264, and *wins* at 1584, exactly the monotone, diameter-driven trend the ballistic mechanism predicts. The four views do not merely fail to contradict one another; they line up along a single axis, the longer the geodesics, the more the $\mathrm{Var}\sim t^2$ reach of the walk matters. That alignment is the signature of a real mechanism rather than a lucky configuration.

The same consistency bounds how far we may push the claim. The scale trend rests on a three-seed budget at 1584 satellites, and at small and medium scale the quantum-inspired operator is at best a tie and is noisier than heat. The honest synthesis for a practitioner is therefore: a global operator is mandatory; the classical heat kernel is an excellent default at small-to-medium diameter; and the quantum-inspired walk is worth adopting where the graph is large and long-range, the regime where our largest-scale experiment shows it is the best operator. That is a more useful and more durable message than a blanket advantage would have been, and it is the message the data actually supports.

IX. How to Evaluate Honestly

The hardest part of quantum-enabled learning is not building the model but trusting the comparison. The wider machine-learning community has learned this lesson the hard way, in reinforcement learning where seeds and implementation details dominate reported gains [64], in generative modeling where a fair large-scale comparison erased claimed advantages [65], and in benchmarking practice more broadly [66], [67]. This section reproduces two pitfalls that manufacture false advantages, shows how to remove them, and distills a checklist. These are not hypothetical: each one produced a misleading signal in our own early experiments before we corrected it.

A. Pitfall 1: scale-coupled normalization

A node-potential target has units of hops, and those units grow with the graph. If we normalize the target by a global graph constant such as the diameter, $y_i = d_G(i, v_d)/\mathrm{diam}(G)$, and then pool errors across graphs of different size, large graphs dominate the average and the apparent global-versus-local gap appears to explode with diameter. Fig. 11 shows the effect: under raw hop units the gap grows with a slope of 0.061 per unit diameter and reaches 0.63 on the largest graph, whereas under a scale-free per-source normalization (by eccentricity) the same gap grows with a slope of only 0.007, roughly 9× flatter, and stays near 0.05.

The honest reading separates two things. A genuine, modest gap remains under the scale-free metric, because a fixed K-hop GCN really does fall behind as the diameter grows. What is spurious is the *magnitude and steepness* of the trend under raw units, which is an artifact of letting the metric scale with the problem. The fix is to normalize per source to a scale-free quantity before pooling heterogeneous graphs, and to report a scale-free summary such as the AURC, in the spirit of standardized robustness protocols [68].

B. Pitfall 2: single-seed reporting

Quantum-inspired models in our study are visibly noisier than their classical counterparts. On the 66-satellite shell over eight seeds, the quantum-inspired walk ranges widely while the heat kernel is tight: 0.085 ± 0.012 versus 0.064 ± 0.005 (the local GCN sits at 0.102 ± 0.010). A single seed could therefore show the quantum-inspired operator beating, matching, or losing to the classical one, as Fig. 12 makes visible. Reporting one run is not a small sin here; it can invert the conclusion. The remedy is to report mean and standard deviation over several seeds and to treat any ranking that lies inside the combined spread as a tie.

C. Pitfall 3: survivorship in the denominator

A subtler trap appears whenever some instances can fail to produce a usable answer, for example a routing decoder that does not deliver every flow, or a method scored only where it converged. Averaging the error over the *surviving* instances

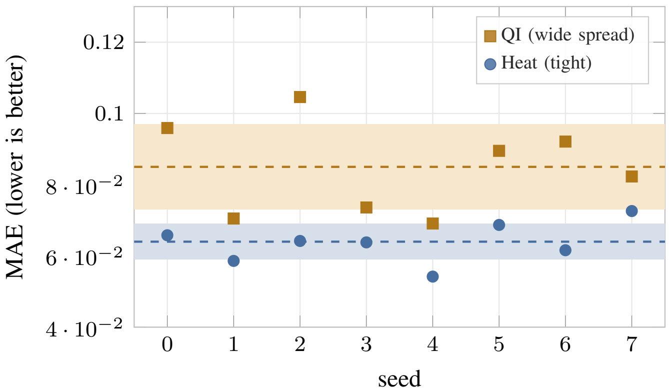

Fig. 12. Exp C2, the single-seed pitfall (66 satellites, 8 seeds). Dashed lines mark the per-operator means and shaded bands the $\pm$ one standard-deviation ranges. The quantum-inspired walk (QI) scatters widely while the heat kernel is tight; a single seed could show QI beating, matching, or losing to heat. Only the distribution, 0.085 ± 0.012 versus 0.064 ± 0.005, is trustworthy.

rewards a method that quietly drops its hardest cases: the harder the case it abandons, the better its conditional mean looks. The fix is to fix the denominator, every instance counts and a dropped case is scored at its true (failed) cost rather than omitted. The same logic applies to seeds and configurations: report all runs, not only the ones that trained cleanly.

D. Pitfall 4: the capacity confound in quantum comparisons

Quantum and classical models are easy to compare unfairly because the quantum knob often moves capacity at the same time. Adding qubits typically widens the classical pre- or post-processing as well, so a quantum advantage measured at matched qubit count can be a plain capacity advantage in disguise. The remedy is to match the resource that actually carries the capacity, the bottleneck width and total parameter count, not the qubit count. We apply this directly: the multi-time quantum-walk variant in Section VIII is given more parameters and is reported with its larger count, so the reader can see it does not buy an improvement despite the extra capacity.

E. A reusable checklist

1) **Param-matched operators.** Vary only the propagation operator; hold depth, width, and weight shapes fixed.
2) **Multiple seeds.** Report mean $\pm$ std; a ranking inside the std is not a result.
3) **Scale-free metrics.** Normalize per source and report a scale-free summary (AURC); never pool raw units across sizes.
4) **Honest denominators.** Count every instance; beware survivorship.
5) **Design sweep before a verdict.** Probe the obvious variants so that "it did not help" is a statement about the mechanism, not one implementation.

X. When to Reach for What

The case study and the operator view together yield a practical decision guide, Table X. The first question is range, not quantumness: if the task needs information to travel farther than a few hops on a large, time-varying graph, a global operator is warranted and a fixed-depth GCN is the wrong tool. Among global operators, the classical heat kernel is a strong, cheap default. The quantum-inspired walk earns its place as the graph grows: ballistic spreading helps most where geodesics are long, so the larger the diameter the more it matters. In our study the quantum-inspired walk trailed heat at 66 satellites, tied at 264, and was the best operator at 1584 (Section VIII-F); a base-matrix sweep (adjacency, Laplacian, normalized Laplacian) further helps on the far-node metric. The practical rule is to prefer the classical heat kernel at small-to-medium scale and to reach for the quantum-inspired walk on large, long-range graphs, adopting it only when a multi-seed comparison confirms a real, not within-noise, gain. True quantum hardware remains a research path rather than a deployment choice today, for the reasons in Section XI.

TABLE X
Operator decision guide. Evidence level reflects this tutorial's study.

Task regime	Recommended operator	Evidence
Short-range, $\leq K$ hops	Local GCN (K layers)	strong
Long-range on large dynamic graph	Classical global (heat $e^{-t\mathbf{L}}$)	strong
Large, long-range graph (big diameter)	Quantum-inspired walk $\lvert e^{-it\mathbf{L}}\rvert^2$ (sweep the base matrix)	best operator at 1584 sats
Exponential feature map needed	True QGNN (research, NISQ-limited)	open

XI. Open Challenges and Roadmap

A. From quantum-inspired to true quantum

The quantum-inspired operator runs on classical hardware because it is a matrix function of the Laplacian. Realizing the walk on quantum hardware promises the exponential-dimensional feature maps a classical model cannot represent, but several obstacles stand in the way. Noisy intermediate-scale quantum (NISQ) devices [24] introduce depolarizing and readout noise that degrades accuracy; qubit counts cap the graph size that can be embedded directly; and parameterized quantum circuits [19] suffer barren plateaus [25] where gradients vanish as the system grows. Quantifying how much of the ballistic mechanism survives device noise is a concrete, near-term experiment: a small parameterized quantum walk on a noisy simulator, swept over depolarizing strength, to measure where the far-node edge of Section VIII erodes. We leave this to future work.

B. Scaling the classical surrogate

Even as a classical surrogate, the eigendecomposition behind $e^{-it\mathbf{L}}$ costs $O(N^3)$, which is acceptable at the hundreds-of-nodes scale of this tutorial but not for full mega-constellations. Krylov-subspace and Chebyshev-polynomial approximations compute $e^{-it\mathbf{L}}\mathbf{x}$ without a full spectrum and are the practical route to thousands of nodes; characterizing the accuracy-cost tradeoff of these approximations for quantum-walk kernels is an open engineering problem.

C. Beyond a single snapshot

LEO topologies are not only dynamic but *predictably* dynamic: the future graph is known from ephemeris. The dynamic-graph methods surveyed in Section IV, TGN [16], EvolveGCN [55], and DySAT [56], learn over a sequence of *past* snapshots and do not use a known future; an operator that propagates over a short horizon of future Laplacians rather than a single snapshot is a promising direction that the snapshot-based study here does not cover.

D. Federated and non-centralized quantum graph learning

In a constellation the graph is physically distributed: each satellite or region holds part of the topology and its local traffic. On-board and cross-satellite federated learning is already an active line for LEO networks [60], [61], building on federated optimization [69]. Running quantum or quantum-inspired graph operators in this setting, where no node sees the whole Laplacian, is open: the spectral operators of this tutorial assume a global eigenbasis, so decentralized or locally-approximated propagation is required. The non-centralized processing of both classical data and quantum states, an explicit theme of this special issue, meets the quantum-internet roadmap here [63].

E. Quantum algorithms for network functions

Beyond learning, quantum search and optimization target network functions directly, for example virtualization and resource placement [62]. How these combine with the learned propagation operators studied here, a learned model proposing candidates that a quantum subroutine refines, is an unexplored but natural hybrid for future wireless management.

F. Benchmarks and reproducibility

Finally, the field needs shared, scale-free benchmarks for graph learning on dynamic wireless topologies, with multi-seed protocols and packet-level validation [38] built in. The pitfalls of Section IX persist largely because evaluation is ad hoc, an observation echoed across machine learning [64], [66], [67]. The companion code to this tutorial is a small step in that direction: every figure is regenerated by a single script from seeded runs.

XII. Threats to Validity

In the spirit of the evaluation discipline this tutorial advocates, we state the limits of our own study so a reader does not over-read it.

Scale and seeds. We test three scales up to 1584 satellites, the largest at which a full eigendecomposition of the Laplacian is affordable on a CPU; beyond it the spectral approximations of Section XI would be required, and they introduce their own error. The headline scale trend, the quantum-inspired walk overtaking the classical kernel as diameter grows (Section VIII-F), rests on three seeds at the 1584 scale because each run is eigendecomposition-heavy; it is monotone and theory-consistent across the three scales, but a larger seed count and still-larger constellations would harden it. The small- and medium-scale results use five and eight seeds and are correspondingly firmer.

Task. We study a single node-potential task chosen to isolate propagation. A different task, for example one with dense rather than single-seed supervision, could shift the balance between local and global operators, and a task whose payoff is not concentrated on far nodes would remove the regime in which ballistic spreading helps. The tutorial's claims are about operators on this task, not a universal ordering.

Snapshot dynamics. Each sample is a single graph snapshot. The predictably dynamic nature of LEO topologies, where future Laplacians are known, is not exploited; operators that propagate over a horizon of future graphs could change the picture and are left to future work.

Quantum-inspired, not quantum. Our quantum-inspired operator runs the quantum-walk kernel on classical hardware. It captures the ballistic-spreading mechanism but not the exponential-dimensional Hilbert-space feature map of a true QGNN, and it does not incur device noise. Conclusions about the quantum-inspired operator therefore bound, but do not settle, what a fault-tolerant quantum implementation would deliver.

XIII. Conclusion

We have presented graph learning for dynamic wireless topologies through a single unifying lens: classical message passing, graph diffusion, and quantum walks are one propagation operator $\mathbf{\Phi}(\mathbf{L})$ with different spreading physics, diffusive versus ballistic. From this lens we derived a param-matched quantum-inspired GNN layer that runs today, and we used a fully reproducible LEO case study to teach not only how to build it but how to judge it. The case study's honest verdict is itself the lesson. Global propagation beats local by a wide margin at every scale; and among global operators the quantum-inspired and classical kernels trade places as the graph grows, the quantum-inspired walk losing at 66 satellites, tying at 264, and becoming the best operator at 1584, the diameter-driven trend its ballistic spreading predicts. The value of a quantum mechanism must be measured, not assumed, and the measurement must survive param-matching, multiple seeds, scale-free metrics, a sweep of the obvious design choices, and a test at the scale where the mechanism is supposed to help. We hope a reader leaves able to place any new quantum graph model in the operator landscape and to evaluate it without fooling themselves.

The practical takeaways are short. (i) Treat GCN, diffusion, and quantum walks as one propagation operator $\mathbf{\Phi}(\mathbf{L})$ and pick the spreading physics the task needs. (ii) On a long-range task over a large dynamic graph, a global operator is mandatory and a fixed K-hop GCN is the wrong tool. (iii) Among global operators, classical heat is a strong, cheap default at small-to-medium scale; reach for a quantum-inspired walk on large, long-range graphs, where our largest-scale experiment shows it becomes the best operator, and confirm any gain with a multi-seed comparison. (iv) True quantum hardware is a research path, not a deployment choice today: the ballistic mechanism is real but fragile to NISQ noise. (v) Above all, evaluate honestly, param-matched operators, multiple seeds, scale-free metrics, honest denominators, and a design sweep before any verdict.

DATA AND CODE AVAILABILITY

All experiments, the LEO topology generator, the operators, and the figure-generation scripts are available at https://github.com/haodpsut/qi-gnn-dynamic-wireless. Every figure in this tutorial is regenerated from seeded runs by the scripts in that repository.

REFERENCES

[1] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2017.

[2] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, 2021.

[3] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, 2017.

[4] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2014.

[5] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2016.

[6] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.

[7] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017.

[8] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[9] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.

[10] M. Chen, Z. Wei, Z. Huang, B. Ding, and Y. Li, "Simple and deep graph convolutional networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020.

[11] J. Gasteiger, A. Bojchevski, and S. Günnemann, "Predict then propagate: graph neural networks meet personalized PageRank," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.

[12] U. Alon and E. Yahav, "On the bottleneck of graph neural networks and its practical implications," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[13] J. Topping, F. Di Giovanni, B. P. Chamberlain, X. Dong, and M. M. Bronstein, "Understanding over-squashing and bottlenecks on graphs via curvature," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[14] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, 2013.

[15] J. Gasteiger, S. Weißenberger, and S. Günnemann, "Diffusion improves graph learning," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.

[16] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," in *ICML Workshop on Graph Representation Learning*, 2020.

[17] S. M. Kazemi, R. Goel, K. Jain, I. Kobyzev, A. Sethi, P. Forsyth, and P. Poupart, "Representation learning for dynamic graphs: a survey," *J. Mach. Learn. Res.*, vol. 21, no. 70, pp. 1–73, 2020.

[18] V. Dunjko and H. J. Briegel, "Machine learning and artificial intelligence in the quantum domain: a review of recent progress," *Rep. Prog. Phys.*, vol. 81, no. 7, p. 074001, 2018.

[19] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, "Variational quantum algorithms," *Nat. Rev. Phys.*, vol. 3, no. 9, pp. 625–644, 2021.

[20] M. Schuld, I. Sinayskiy, and F. Petruccione, "The quest for a quantum neural network," *Quantum Inf. Process.*, vol. 13, no. 11, pp. 2567–2586, 2014.

[21] M. Schuld and N. Killoran, "Quantum machine learning in feature hilbert spaces," *Phys. Rev. Lett.*, vol. 122, no. 4, p. 040504, 2019.

[22] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.

[23] A. Abbas, D. Sutter, C. Zoufal, A. Lucchi, A. Figalli, and S. Woerner, "The power of quantum neural networks," *Nat. Comput. Sci.*, vol. 1, no. 6, pp. 403–409, 2021.

[24] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.

[25] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nat. Commun.*, vol. 9, no. 1, p. 4812, 2018.

[26] G. Verdon, T. McCourt, E. Luzhnica, V. Singh, S. Leichenauer, and J. Hidary, "Quantum graph neural networks," *arXiv:1909.12264*, 2019.

[27] P. Mérnyei, K. Meichanetzidis, and İ. İ. Ceylan, "Equivariant quantum graph circuits," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2022.

[28] L.-P. Henry, S. Thabet, C. Dalyac, and L. Henriet, "Quantum evolution kernel: machine learning on graphs with programmable arrays of qubits," *Phys. Rev. A*, vol. 104, no. 3, p. 032416, 2021.

[29] E. Farhi and S. Gutmann, "Quantum computation and decision trees," *Phys. Rev. A*, vol. 58, no. 2, pp. 915–928, 1998.

[30] J. Kempe, "Quantum random walks: an introductory overview," *Contemp. Phys.*, vol. 44, no. 4, pp. 307–327, 2003.

[31] A. M. Childs, R. Cleve, E. Deotto, E. Farhi, S. Gutmann, and D. A. Spielman, "Exponential algorithmic speedup by a quantum walk," in *Proc. ACM Symp. Theory Comput. (STOC)*, 2003, pp. 59–68.

[32] A. M. Childs and J. Goldstone, "Spatial search by quantum walk," *Phys. Rev. A*, vol. 70, no. 2, p. 022314, 2004.

[33] A. M. Childs, "Universal computation by quantum walk," *Phys. Rev. Lett.*, vol. 102, no. 18, p. 180501, 2009.

[34] S. E. Venegas-Andraca, "Quantum walks: a comprehensive review," *Quantum Inf. Process.*, vol. 11, no. 5, pp. 1015–1106, 2012.

[35] R. Portugal, *Quantum Walks and Search Algorithms*. Springer, 2013.

[36] O. Kodheli, E. Lagunas, N. Maturo, S. K. Sharma, B. Shankar, J. Q. Montoya, J. C. M. Duncan, D. Spano, S. Chatzinotas, S. Kisseleff *et al.*, "Satellite communications in the new space era: a survey and future challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 70–109, 2021.

[37] M. Handley, "Delay is not an option: low latency routing in space," in *Proc. ACM Workshop Hot Topics Netw. (HotNets)*, 2018, pp. 85–91.

[38] S. Kassing, D. Bhattacherjee, A. B. Águas, J. E. Saethre, and A. Singla, "Exploring the "internet from space" with hypatia," in *Proc. ACM Internet Meas. Conf. (IMC)*, 2020, pp. 214–229.

[39] K. Rusek, J. Suárez-Varela, P. Almasan, P. Barlet-Ros, and A. Cabellos-Aparicio, "RouteNet: leveraging graph neural networks for network modeling and optimization in SDN," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 10, pp. 2260–2270, 2020.

[40] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, "Graph neural networks for scalable radio resource management: architecture design and theoretical analysis," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 101–115, 2021.

[41] M. Eisen and A. Ribeiro, "Optimal wireless resource allocation with random edge graph neural networks," *IEEE Trans. Signal Process.*, vol. 68, pp. 2977–2991, 2020.

[42] H. Mao, M. Alizadeh, I. Menache, and S. Kandula, "Resource management with deep reinforcement learning," in *Proc. ACM Workshop Hot Topics Netw. (HotNets)*, 2016, pp. 50–56.

[43] J. Zhou, G. Cui, S. Hu, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, and M. Sun, "Graph neural networks: a review of methods and applications," *AI Open*, vol. 1, pp. 57–81, 2020.

[44] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, "Geometric deep learning: going beyond euclidean data," *IEEE Signal Process. Mag.*, vol. 34, no. 4, pp. 18–42, 2017.

[45] A. Ortega, P. Frossard, J. Kovačević, J. M. F. Moura, and P. Vandergheynst, "Graph signal processing: overview, challenges, and applications," *Proc. IEEE*, vol. 106, no. 5, pp. 808–828, 2018.

[46] F. R. K. Chung, *Spectral Graph Theory*. American Mathematical Society, 1997.

[47] D. K. Hammond, P. Vandergheynst, and R. Gribonval, "Wavelets on graphs via spectral graph theory," *Appl. Comput. Harmon. Anal.*, vol. 30, no. 2, pp. 129–150, 2011.

[48] Q. Li, Z. Han, and X.-M. Wu, "Deeper insights into graph convolutional networks for semi-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2018.

[49] S. Dernbach, A. Mohseni-Kabir, S. Pal, M. Gepner, and D. Towsley, "Quantum walk neural networks with feature dependent coins," *Appl. Netw. Sci.*, vol. 4, 2019.
[50] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, "Quantum circuit learning," *Phys. Rev. A*, vol. 98, no. 3, p. 032309, 2018.
[51] E. Farhi and H. Neven, "Classification with quantum neural networks on near term processors," *arXiv:1802.06002*, 2018.
[52] M. Schuld and F. Petruccione, *Supervised Learning with Quantum Computers*. Springer, 2018.
[53] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, and J. R. McClean, "Power of data in quantum machine learning," *Nat. Commun.*, vol. 12, no. 1, p. 2631, 2021.
[54] Y. Liu, S. Arunachalam, and K. Temme, "A rigorous and robust quantum speed-up in supervised machine learning," *Nat. Phys.*, vol. 17, no. 9, pp. 1013–1017, 2021.
[55] A. Pareja, G. Domeniconi, J. Chen, T. Ma, T. Suzumura, H. Kanezashi, T. Kaler, T. Schardl, and C. Leiserson, "EvolveGCN: evolving graph convolutional networks for dynamic graphs," in *Proc. AAAI Conf. Artif. Intell.*, 2020.
[56] A. Sankar, Y. Wu, L. Gou, W. Zhang, and H. Yang, "DySAT: deep neural representation learning on dynamic graphs via self-attention networks," in *Proc. ACM Int. Conf. Web Search Data Min. (WSDM)*, 2020.
[57] J. Suárez-Varela, P. Almasan, M. Ferriol-Galmés, K. Rusek, F. Geyer, X. Cheng, X. Shi, S. Xiao, F. Scarselli, A. Cabellos-Aparicio, and P. Barlet-Ros, "Graph neural networks for communication networks: context, use cases and opportunities," *IEEE Network*, vol. 37, no. 3, pp. 146–153, 2023.
[58] W. Jiang, "Graph-based deep learning for communication networks: a survey," *Comput. Commun.*, vol. 185, pp. 40–54, 2022.
[59] I. del Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low earth orbit satellite constellation systems to provide global broadband," *Acta Astronaut.*, vol. 159, pp. 123–135, 2019.
[60] N. Razmi, B. Matthiesen, A. Dekorsy, and P. Popovski, "On-board federated learning for dense LEO constellations," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2022.
[61] B. Matthiesen, N. Razmi, I. Leyva-Mayorga, A. Dekorsy, and P. Popovski, "Federated learning in satellite constellations," *IEEE Network*, vol. 38, no. 2, pp. 232–239, 2024.
[62] P. Botsinis, D. Alanis, Z. Babar, H. V. Nguyen, D. Chandra, S. X. Ng, and L. Hanzo, "Quantum search algorithms for wireless communications," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 2, pp. 1209–1242, 2019.
[63] S. Wehner, D. Elkouss, and R. Hanson, "Quantum internet: a vision for the road ahead," *Science*, vol. 362, no. 6412, p. eaam9288, 2018.
[64] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep reinforcement learning that matters," in *Proc. AAAI Conf. Artif. Intell.*, 2018.
[65] M. Lucic, K. Kurach, M. Michalski, S. Gelly, and O. Bousquet, "Are GANs created equal? a large-scale study," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2018.
[66] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research," *J. Mach. Learn. Res.*, vol. 22, no. 164, pp. 1–20, 2021.
[67] M. Dehghani, Y. Tay, A. A. Gritsenko, Z. Zhao, N. Houlsby, F. Diaz, D. Metzler, and O. Vinyals, "The benchmark lottery," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.
[68] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
[69] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*, 2017.

Phuc Hao Do is a Lecturer and Researcher in the Faculty of Information Technology at Danang Architecture University (DAU), Viet Nam. He received the Ph.D. degree in telecommunications systems and networks from the Bonch-Bruevich Saint Petersburg State University of Telecommunications (SPbSUT), Russia, in 2026. His research interests include intelligent 6G non-terrestrial networks, quantum-inspired AI, Kolmogorov-Arnold networks (KAN), resilient computing, and network security. He has authored peer-reviewed papers in international journals and conferences on IoT malware detection, satellite communication optimization, and deep learning architectures.